

View-Adaptive Asymmetric Image Detail Enhancement for 360-degree Stereoscopic VR Content

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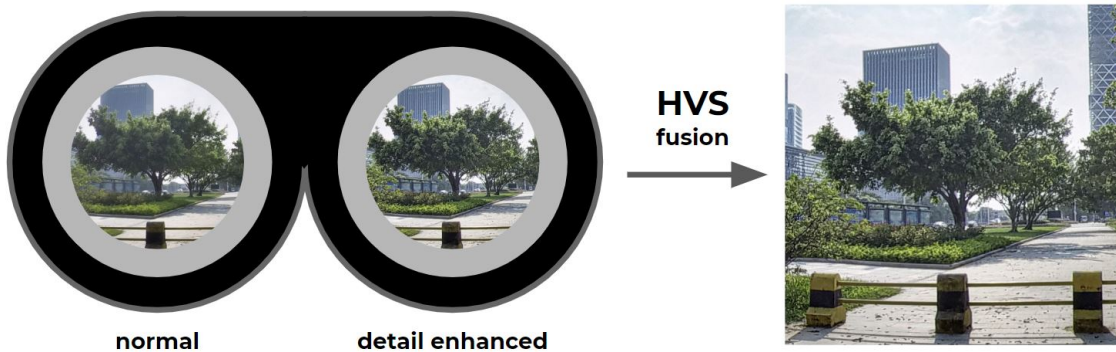


Figure 1: *Asymmetric Image Detail Enhancement*. The enhancement process is only introduced to one single channel and the high-frequency details shall dominate the final percept according to the suppression theory of binocular vision. [1,4]

ABSTRACT

We present a simple VR-specific image detail enhancement method that improves the viewing experience of 360-degree stereoscopic photographed VR contents. By exploiting the fusion characteristics of binocular vision, we propose an asymmetric process that applies detail enhancement to one single image channel only. Our method can dynamically apply the enhancement in a view-adaptive fashion in real-time on most low-cost standalone VR headsets. We discuss the benefits of this method with respect to authoring possibilities, storage and bandwidth issues of photographed VR contents.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Computer graphics—Image manipulation—Image processing;

1 INTRODUCTION

Propelled by the recent success of standalone virtual reality headsets, the demand for immersive content is rising rapidly. Real-time computer graphics-based content is often expensive and time-consuming to produce, photographed 360-degree stereo images and videos have inevitably become a significant source of immersive VR content.

Most 360-degree cameras are equipped with multiple image sensors, and the images they capture are stitched together to produce a single image of very high resolution that is far beyond what a single sensor can afford. However, the inconsistent exposure or white balance between the sensors might lead to a stitched image that exhibits various unevenness in image quality. Therefore these stereoscopic 360-degree images often require careful post-processing for maximising the global image quality. Since this process is carried out globally, these 360-degree stereoscopic images are often treated as regular images without considering the VR-specific viewing factors.

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When we carefully consider how 360-degree stereoscopic images are experienced through a VR headset, we discover two VR-specific characteristics that we can exploit to improve the viewing experience. These two unique features are namely the *Limited Field of View* and *Binocular Vision*.

Limited Field of View. The limited field of view of VR headsets naturally limits the portion that we can see from these 360-degree images. However, we may transform this limit into an opportunity for image enhancement in a view-dependent manner. We propose a view-adaptive enhancement to the local contrast of the visuals that the viewer is seeing. Hence, the degree of detail enhancement can be made adaptive and dynamically controlled by the headset's orientation relative to the 360-degree image.

Binocular Vision. VR headsets deliver stereoscopic contents by streaming two different images to our left and right eyes to produce a stereo percept via our binocular vision. Such independence between the left and right eye channels offers a second possibility for enhancing the viewing experience. Video engineers have explored how a human vision system can accommodate asymmetric image quality in a stereo setup [8]. They found that mixed-resolution stereoscopic contents can often reliably produce a binocular percept for most viewers. The perceived single image sharpness is comparable to the higher resolution image channel of the asymmetric content. This discovery led to applying spatial filtering to one image channel of stereo contents to reduce the total transmission bandwidth.

The above asymmetric approach agrees with the suppression theory of binocular vision [1] and this specific case of suppression where the high-resolution member of an asymmetric stereo image pair dominates the final percept has been explicitly described in a previous perception research paper [4]. Inspired by this suppression theory and other binocular vision-based methods in the computer graphics area, we propose an asymmetric image detail enhancement approach that improves the quality of the binocular percept of a stereo image pair by only enhancing a single image channel. We have carried out a small user study to evaluate the subjectively perceived quality of the stereo images produced by our asymmetric method.

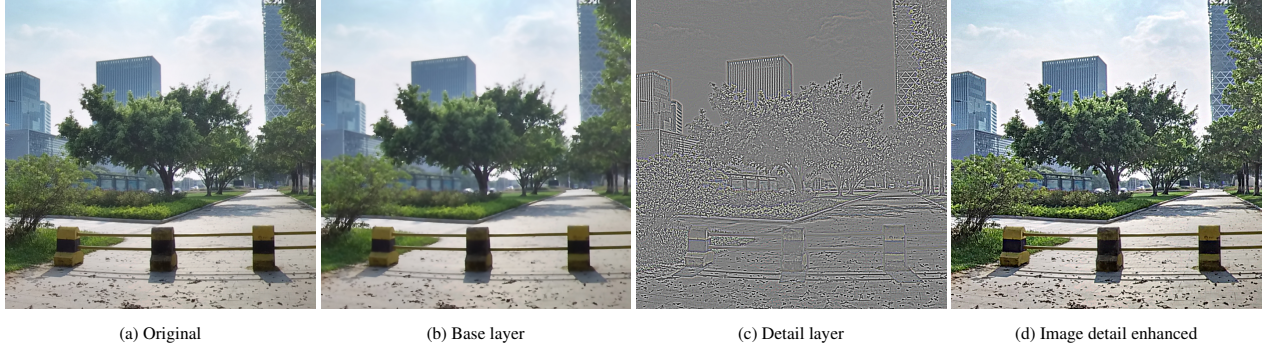


Figure 2: Decomposition and enhancement example using the sub-window variance based decomposition method [10]

By coupling the view adaptation and an asymmetric detail enhancement process, our method introduces a novel way to enhance 360-degree stereoscopic content specifically for Virtual Reality. The main contributions of our work are as follows:

- Our view-adaptive method enables novel VR content authoring possibilities such as view angle or content-sensitive enhancement.
- The proposed asymmetric detail enhancement process requires less computation and data storage while maintaining an improved viewing experience.

2 RELATED WORK

Asymmetric methods are standard in the area of stereo video coding. There is a method [6] that proposes to degrade the chrominance information of the right eye image channel, and it results in no noticeable degradation of the overall experience. An alternative video coding method [2] filters asymmetric alternate slices in each eye channel such that there is a minimal visual difference between the two channels. The fusion characteristic of binocular vision enables many innovative applications. It is possible to use stereoscopic content as a means to share images with people who have colour vision deficiency [7]. Monoscopic HDR image viewing on stereo displays can be enhanced by binocular tone mapping [11].

In terms of view-dependent VR image enhancement, there is a method that turns the tone mapping operator into a view adaptive one [5]. The potential benefit of this approach is obvious, but the proposed linear tone-mapping approach may not fully reveal the advantage of view-adaptive tone mapping for VR.

3 ASYMMETRIC IMAGE DETAIL ENHANCEMENT

This section describes our asymmetric image detail enhancement process, followed by a simple user study that evaluates the subjective perception quality of the images produced by our method.

3.1 Image Detail Enhancement

To improve a given stereo pair's image detail, we propose applying the enhancement process only to one single image channel, either the left or right-eye image. Image detail manipulation requires iterative application of a robust edge-preserving filter [3, 9] to decompose an image into multiple components of different levels of detail. The following equation describes this form of decomposition:

$$I = B + \sum_{i=1}^N D_i \quad (1)$$

where I is the original image, and it is decomposed into a base layer B and multiple detail layers D_i . The base layer is often a piecewise smooth image that maintains the low-frequency information and the primary image structures that define the overall dynamic range of the original image. The detail layers D_i hold the image detail of different scales. The enhancement process is thus achieved by increasing the contribution of each detail component in the final image.

In our method, we use a modern edge-preserving filter named *sub-window variance filter* [10] for decomposition and image detail manipulation. Fig. 2 shows an example of the enhancement result. This filter uses two intuitive filter parameters, namely the filter radius and an image detail threshold parameter ϵ . The value of ϵ ranges from 0 to 0.25 for LDR images, and it defines the minimum threshold variance of image details to be filtered. Due to the limited space of this paper, we refer our readers to the original article [10] for the filter's formulation and its implementation detail.

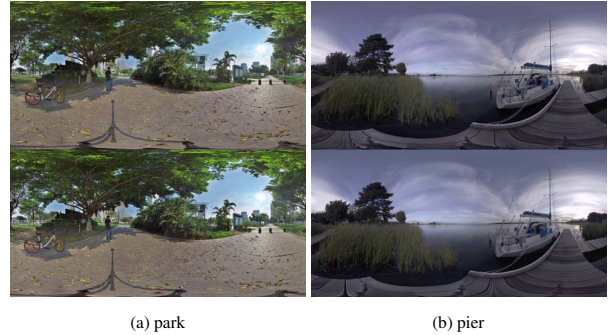


Figure 3: Asymmetrically enhanced 360-degree stereo image pairs used in our experiments. (a) the 'park' stereo pair has a resolution of 7680 $\times$ 7680 with right eye channel detail enhanced. (b) the 'pier' stereo pair has a resolution of 3840 $\times$ 3840 with left eye channel detail enhanced.

3.2 Subjective Perception Experiment

We have prepared two groups of 360-degree stereo image pairs in the top-bottom lat-long image format. There are three stereo image pairs in each group; the original, the asymmetrically enhanced and the fully enhanced (both left and right images enhanced). In the 'park' group, the asymmetrical enhancement was applied to the right-eye

image channel. In the 'pier' group, the enhancement was applied to the left-eye image. Fig. 3 shows the asymmetrically enhanced image pairs.

The *fine detail* and *medium detail* layers of both test sets were extracted using the sub-window variance filter [10] with an identical set of filter parameters. The fine detail layers were extracted with a filter radius of 2 and $\varepsilon = 0.01$; the medium layers were extracted with a filter radius of 8 and $\varepsilon = 0.02$. However, the degrees of detail enhancement applied to the two groups were different, and are listed in Table 1. The 'park' stereo pair was more aggressively enhanced, and the enhancement applied to the 'pier' set was mild (only the fine detail layer was enhanced).

Table 1: Detail enhancement weights

Image Pair	Fine detail weight	Medium detail weight
Park	3.0	2.0
Pier	1.5	1.0

We invited ten individuals (age from 19 to 55) to participate in this experiment. All of them used the same Oculus Quest headset to view the stereo pairs (this headset has a native resolution of 1440×1600 per eye). We asked each subject to evaluate and prioritize the perceived level of details of each stereo pair by assigning a score from 1 to 3 points (the *Best* = 3 points, the *Good* = 2 points, and the *Fair* = 1 point). The evaluation results are shown in Fig. 4 and Fig. 5.

In the case of the 'park' image pair, the rating of the asymmetrically enhanced one is high. The result suggests that the asymmetric image pair is better than the original to most subjects. In addition, the asymmetric image pair almost matches the perceived quality of the fully enhanced image pair. For the mildly enhanced test set, i.e. the 'pier' set, the fully enhanced image pair performs the best, and the asymmetric image pair set serves better than the original one. It seems our asymmetric approach works better when the degree of detail enhancement is high.

These early experiments provide positive results, but we believe that additional experiments in terms of image genre variety and degree of detail enhancements should be conducted in future evaluations.

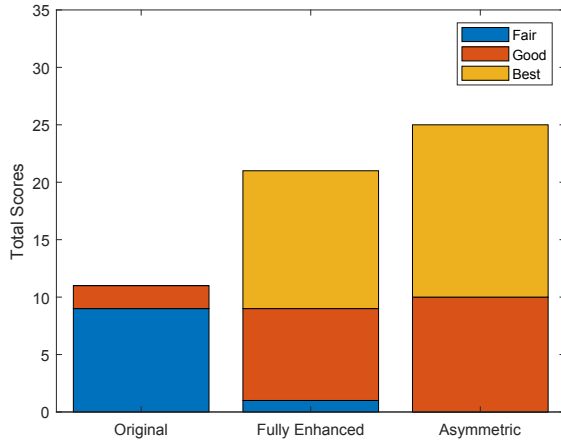


Figure 4: viewing scores of the 'Park' image pairs

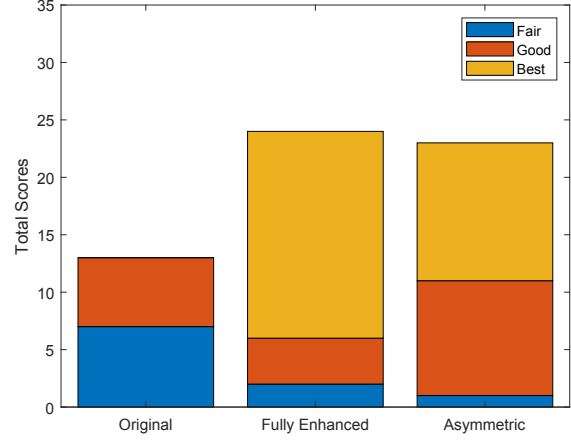


Figure 5: viewing scores of the 'Pier' image pairs

4 VIEW-ADAPTIVE ENHANCEMENT

To integrate view-dependence into our image detail enhancement process is straight forward. We propose to pre-decompose the target images into its base and detail layer(s), and we dynamically composite them together on the VR headset hardware using a set of weight factors defined by the current pan-and-tilt angles of the headset. The following equation describes the compositing process:

$$I_v = B + \sum_{i=1}^N w_i(\phi, \lambda) D_{iv} \quad (2)$$

where B is the base layer and I_v is the resultant image displayed by the headset. D_{iv} are the visible regions of the detail layers determined by the viewing angle. $w_i(\phi, \lambda)$ is the view-adaptive weight function; ϕ and λ are the pan and tilt angles, i.e. the viewing angle of the VR headset. There are many ways we can define this weight function $w_i(\phi, \lambda)$, such as using a lookup texture map parameterised by ϕ and λ or simple procedural weight functions.

In our implementation, we use a simple weight function that linearly scales with the viewing angle values. The weight function is implemented using *glsl* shader language, and the view-adaptive enhancement pipeline runs reliably in real-time, i.e. 60 fps on the Oculus Quest headset. We have included a short demo video (captured in a Oculus Quest VR session) in our supplemental material; the video demonstrates how the viewing angle controls the degree of image detail enhancement. Fig. 6 shows two selected frames from the video.

5 DISCUSSION

The main takeaway of our proposed method is the possibility of applying the image detail enhancement process to only one single image of a stereo pair. From a pragmatic point of view, our approach allows budget-sensitive productions to eliminate the need of resolving correspondence between the left and right eye channels when only certain selected regions require detail enhancement.

Similar to the findings of most asymmetric video coding methods, our asymmetric process reduces the amount of high-frequency information by 50% when compared with a complete enhancement process. Table 2 shows the JPEG file size of the image pairs used in our test sets. In short, our asymmetric enhancement approach offers the advantage of less computation, reduced storage and hence bandwidth.



Figure 6: The video frames captured from our demo video showing how the degree of enhancement is made view-adaptive.

Table 2: Stereo pair JPEG image size (in MB)

Image types	park (8K)	pier (4K)
Not enhanced	49.8	8.7
Asymmetrically enhanced	65.2	10.1
Fully enhanced	80.6	11.4

6 FUTURE WORK

As mentioned before, we believe additional subjective perception studies will be necessary to identify the limits and the preferred conditions of using this asymmetric approach. We see another potential application of our method is to create an in-VR stereoscopic image enhancement application where the author can carefully design the view-dependent weight function in VR. Lastly, we are planning to extend our method to moving images. However, we are aware that a robust temporal coherent edge-preserving filter will be required for such purpose.

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